Jonathan Rufus Samuel

Systems Software Engineer | Systems Design & Software Architecture | Open-Source Advocate

PROFESSIONAL & RESEARCH EXPERIENCE

CERN | Software Infrastructure & DevOps Engineer

September 2024 – Present | Geneva, Switzerland

• DevOps Engineer for Windows - Work on Systems Administration for Windows, DevOps Engineering for Sysadmin services and Infrastructure Software Development.
• Spearheaded research and development of **Package Automator** - a Provisioning via code CI/CD tool that helps automatically create, test and deploy packages across both Intune and MECM. Cut down day to day tasks of team from over 7 days of work each month to 1 hour. Set to be presented as **Next-gen tools for the new era of Windows device management at CERN** at HEPiX Fall Conference 2026, Lincoln, USA.
• Aided the multi-year migration from in house management software for over 10,000 machines to hybrid industry backed solutions - Microsoft's MCM and Intune.
• Revamped the Email Alert's Web App using .Net Core, which currently serves for sending email alerts for over 10,000+ machines on site, as well as developed a fully configurable notifications service with a simple compiler to dynamically interpret configurations into code - Custom Notification template forms for administrators.
Tools Used: C#, Python, React, Git, OpenShift, OpenStack, Docker, Azure, Kubernetes, MSSQL, Influx DB, Grafana and Microsoft Configuration Manager and Intune.

HYPERFACE | SDE Intern - Backend Engineering

January 2024 – July 2024 | Bangalore, India

• Led a team of 3 Application Engineers to complete company wide APIGEE updation and migration, spanning over 4 microservices within the Hyperface application suite.
• Wrote Python Scripts for migrating over 2 million bank records to our servers for further data processing and analysis from various Banking providers.
• Designed and developed a centralized Cron Tracker Mechanism within our orchestrator service and developed the end-to-end functionality for AU Bank's Virtual Card Handling feature, whilst leading a team of 3 Application Engineers.
Tools Used: Java, Apache Groovy, Java Spring Boot, Spring Security, Apache Kafka, Gradle, Postman, MySQL, Python Scripting, Burpsuite.

VIT VELLORE | Student Research & Developer

February 2024 – August 2024 | Bangalore, India

• Worked with Shivansh Sahai and Dr. Swarnlatha P to create **Automated Mp3 Tag Editor**, a Rust based desktop app that scrapes metadata of music tracks from various sources, and classifies them via vector based indexing for local maintenance. Published as a chapter in **Handbook of Research on Deep Learning Techniques for Cloud-Based Industrial IoT by IGI Global**. DOI: 10.4018/978-1-6684-8098-4.ch012.
• Worked with Jaskaran Singh, Shreyash Mehrotra and Dr. Baiju B. V. to research on the simulation challenges faced by IoT developers in the field of Fog and Edge Computing, and how to overcome such simulation challenges via open-source development. Published in proceedings of **2023 International Conference on Next Generation Electronics (NEleX)**. DOI: IEEEXplore ISBN NO: 979-8-3503-1908-8
Tools Used: Rust, Python, FastApi, React, Tauri, Azure, Docker, Sqlite, ShadCN UI.

PROJECTS & OPEN-SOURCE WORK

DIJKSTRA | Founder, Chief Maintainer

February 2023 – present | Bangalore | github.com/Dijkstra-Edu

• Started an open-source organization that helps students prepare for careers in technology through proof-of-work-based open-source projects.
• Led design and student-run development of various projects and services including a latex resume builder, a blog mechanism for students, a GitHub statistics aggregator and a custom Task Tracker, spread across 20 repositories and multiple frameworks.
• Actively mentored over 50 students on software development and best practices, as well as maintaining documentation, software infrastructure and deployment lifecycles.
Tools Used: TypeScript, Next.Js, ShadCN UI, Java, SpringBoot, Rust, Axum, Go, Gin, Node.js, Express.js, JavaScript, Astor, Azure Cloud, AWS, GCP, Docker, Kubernetes, Git & Github, Supabase, MongoDB, PostGreSQL, Postman.

EDUCATION

VIT UNIVERSITY, VELLORE

B.Tech Computer Science Engineering w/ Specialization in IoT
CGPA: 8.51
2020-2024| Vellore, India

BISHOP COTTON BOYS' SCHOOL

2005-2020| Bangalore, India
12th Boards: 93.5% (ISC)
10th Boards: 91.2% (ICSE)

ACHIEVEMENTS

• Completed **Microsoft Azure Administrator Associate (AZ-104)** & **Azure Fundamentals (AZ-900)** Certification offered by **Microsoft**.
• Former **Beta Level Student Learn Ambassador (MLSA Program)** at **Microsoft**.
• Open Source Contributor at **AppFlowy**, **CERN Root**, **Dijkstra**, **Apprentice** and **YAFS**.
• Mathematics for Machine Learning Specialization offered by **Imperial College London** & Completed Accelerated Computer Science Fundamentals Specialization offered by **University of Illinois (GoldmanSachs ECHP-23)**
• Attended the CERN school of computing 2026 held in Latin America, Santiago Chile at the **Pontificia Universidad Católica de Chile (UC)**. Earned a Diploma in Physics Computing, Software Engineering & Data Technologies.
• CERN Guide for the Atlas Experiment, Anti-Proton Decelerator and Data Center.
• Mentor at CERN and Dijkstra to student community, ex-mentor to IEEE-TEMS

SKILLS

Programming

TypeScript & Javascript • Java • Python • C# • Rust • Go • C++ • SQL (MySQL, PostgreSQL, MSSQL & SQLite3) • C/C++ • Groovy

Frameworks & Technologies

Next.Js • .Net Core • Spring Boot • FastAPI • React JS • Axum • Gin • Git & GitHub, Gitlab • NodeJS & ExpressJS • Azure Cloud • AWS Cloud • MongoDB • Supabase • REST API Design • GraphQL • Docker • Kubernetes • OpenShift • OpenStack • Compiler Design • Edge Computing • Tauri • Selenium • Jest • Linux • Scrum & Agile Methods • DevOps • Systems Design • Intune & MECM

LINKS

Github: https://github.com/JRS296
LinkedIn: https://linkedin.com/in/jrs2002/
Website: https://jrs296.github.io/JRS-Studios/
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